LASERS

I. INTRODUCTION

LASER is the acronym of *LIGHT AMPLIFICATION BY STIMULATED EMISSION OF RADIATION*. Laser is one of the out standing inventions of the 20^{th} century in the field of applied optics. A laser beam is a highly parallel coherent beam of light of very high intensity. Laser action is achieved by creating population inversion which can be achieved by different means, such as optical pumping, electrical discharge ….etc., The first ever known laser was invented by T.H.MAIMAN in the year 1960. The theory behind this was given by EINSTEIN in the year 1916.

II. PRINCIPLE AND PRODUCTION OF LASERS

Radiation interacts with matter under appropriate conditions .The interaction leads to an abrupt transition of the quantum system such as atom or an molecule from one energy state to another. If the transition is from lower energy state to higher energy state ,the system absorbs the incident energy. If the transition is from higher energy state to lower energy state,then the system gives out a part of its energy.

Consider two energy states E_1 and E_2 of a system. If the energy diference between the two energy levels is ΔE, then $\Delta E = (E_2 - E_1)$ -- (1)

According to Max Planck ,if an electromagnetic radiation of frequency

$$\gamma = \Delta E/h = (E_2 - E_1)/h \quad \text{-- (2)}$$

is incident on the system which is in the energy state E_1,then the system can move to the energy state E_2 by absorbing the energy.On the other hand ,if a system in state E_2 emits an em radiation of frequency γ given by Equation(2),then the energy of the system changes to E_1.

There are three possible ways through which interaction of radiation and matter can take place, absorption of radiation is one means,known as induced absorption.Two ways of emission namely the spontaneous emission and the stimulated emission.

1. Induced Absorption

Induced absorption is the absorption of an incident photon by a system as a result of which the system is elevated from a lower energy state to an higher energy state.

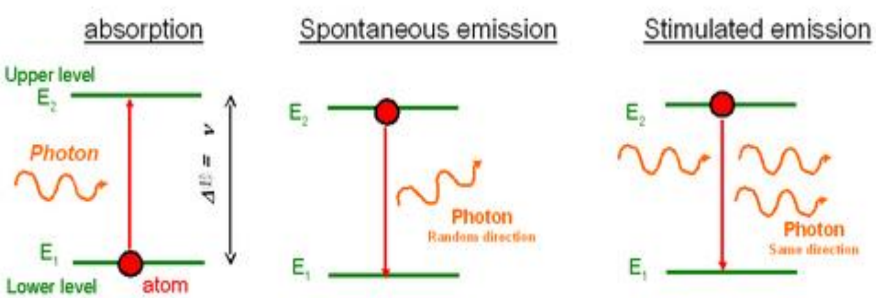


Figure-1

Let E_1 and E_2 be two energy levels in the energy level scheme of an atom in which E_1 corresponds to lower energy as in figure-1. Let us assume that the atom is in the lower energy state (ground state). Let a photon having an energy $\Delta E = (E_2 - E_1)$ be incident on the atom under which condition,the atom absorbs the photon. As a result, its energy becomes $E_1 + \Delta E = E_2$. Now the atom is said to have made transition to the excited state and is indicated as atom*.

This is called induced absorption and could be represented as,

$$\text{atom} + \text{photon} \text{ ------------> } \text{atom}^*$$

2. **Spontaneous Emission**

Spontaneous emission is the emission of a photon, when a system transits from a higher energy state to a lower energy state without the aid of any external agency.

Consider an atom in the excited state as shown in figure-1. Because of the universal tendency for any system to attain the least available energy state, the atom voluntarily emits a photon of energy $\Delta E = (E_2 - E_1)$. The energy of the atom then becomes equal to $E_2 - \Delta E = E_1$.

Since the atom emits the photon without being aided by any external means, it is called spontaneous emission.

The photon may be emitted in any direction. Two such photons which are spontaneously emitted by two atoms under identical conditions may not have any phase similarities and they may not come in same direction. Hence they are incoherent. The process can be denoted as,

$$\text{atom}^* \text{ ---------------> } \text{atom} + \text{photon}$$

3. **Stimulated Emission**

Stimulated emission is the emission of a photon by a system under the influence of a passing photon of just the right energy, due to which the system transits from higher energy state to a lower energy state as shown in figure-1. The photon thus emitted is called the stimulated photon and will have the same phase energy and direction of movement as that of the passing photon called the *stimulating photon*.

Consider an atom in the excited state. Let a photon having an energy $\Delta E = (E_2 - E_1)$ interact with the atom by passing in its vicinity. Under such stimulation, the atom emits a photon and transits to the lower energy state. The two photons travel in exactly the same direction, with same energy and they have identical phase. Thus they are coherent. This process can be denoted as,

$$\text{atom}^* + \text{photon} \text{ ---------------> } \text{atom} + \text{photon} + \text{photon}$$

This kind of emission is responsible for laser action.

Boltzmann factor

The population of different energy states of any physical system are related to each other, provided the system is in thermal equilibrium. Among the various energy states, if we consider any two energy states E_1 and E_2 with population N_1 and N_2 respectively, and $E_2 > E_1$, then the Boltzmann's factor is the ratio (N_2 / N_1) is given by:

$$\frac{N_2}{N_1} = e^{-h\gamma/kT}$$

Where k is the Boltzmann's constant

Since $E_2 > E_1$, $e^{-h\gamma/kT} < 1$

Therefore $N_2 < N_1$

Hence for a system in thermal equilibrium, the population of any higher energy state (N_2) will always be lesser than that in any of the lower energy states (N_1).

III. EXPRESSION FOR ENERGY DENSITY IN TERMS OF EINSTEIN'S COEFFICIENTS

Conside two energy states E_1 and E_2 of a system of atoms.Let there be N_1 atoms with energy E_1 and N_2 atoms with energy E_2 per unit volume of the system. N_1 and N_2 are called the number density of atoms in the states 1 and 2 respectively.Let there be radiation of frequency γ such that $\gamma = (E_2 - E_1) / h$, and let $U\gamma$ be the energy density of radiation of frequency γ. Then $U\gamma d\gamma$ will be the energy density of radiation whose frequencies lie in the range γ *and* $\gamma + d\gamma$.

1.Induced absorption

In case of induced absorption, an atom in the level E_1 can go to level E_2 when it absorbs a radiation of frequency γ .The number of such absorption per unit time per unit volume is called rate of absorption.

The rate of absorption depends upon,

a) The number density of lower energy state,i.e, N_1 and,

b) The energy density i.e,

Therefore, the rate of absorption $\propto N_1U\gamma$

Or the rate of absorption $= B_{12}N_1U\gamma$ → (1)

Where, B_{12} is the constant of proportionality called Einstein's coefficient of induced absorption.

2. Spontaneous emission

In this case, an atom in the higher energy state E_2 undergoes transition to the lower energy state E_1 voluntarily by emitting a photon. Since it is a voluntary transition, it is independent of the energy density of any frequency in the incident radiation. The number of such spontaneous emission per unit time per unit volume,is called rate of spontaneous emission which is proportional to only the number density in the higher energy state, i.e, N_2.

Therefore, the rate of spontaneous emission $\propto N_2$

Or the rate of spontaneous emission $= A_{21} N_2$ → (2)

Where, A_{21} is the constant of proportionality called Einstein's coefficient of spontaneous emission.

3. Stimulated emission

In this case the system requires an external photon of frequency γ, to stimulate the atom for the corresponding downward transition, and there by causing emission of stimulated photons. The number of such stimulated emissions per unit time per unit volume is called rate of stimulated emission which is proportional to,

a) The number density of the higher energy state i.e., N_2 and

b) The energy density , i.e , $U\gamma$.

Therefore, the rate of stimulated emission $\propto N_2 U\gamma$.

Or, the rate of stimulated emission = $B_{21} N_2 U\gamma$. → (3)

Where B_{21} is the constant of proportionality called Einstein coefficient of stimulated emission.

EXPRESSION FOR ENERGY DENSITY:

Let the system be in thermal equilibrium which means that, the total energy of the system remains unchanged inspite of the interaction that is taking place between itself and the incident radiation. Under such a condition, the number of photons absorbed by the system per second must be equal to the number of photons it emits per second by both the stimulated and the spontaneous emission processes.

Therefore at thermal equilibrium,

Rate of absorption = Rate of spontaneous emission + Rate of stimulated emission.

Therefore from Eqs(1), (2) & (3) we can get,

$$B_{12}N_1U\gamma = A_{21}N_2 + B_{21}N_2U\gamma$$

$$U\gamma[B_{12}N_1 - B_{21}N_2] = A_{21}N_2$$

$$U\gamma = \frac{A_{21}N_2}{[B_{12}N_1 - B_{21}N_2]}$$

By rearranging the above equation, we get,

$$U\gamma = \frac{A_{21}N_2}{B_{21}N_2\left[\frac{B_{12}N_1}{B_{21}N_2} - 1\right]}$$

$$U\gamma = \frac{A_{21}}{B_{21}\left[\frac{B_{12}N_1}{B_{21}N_2} - 1\right]}$$

But Boltzmann's law, we have $\frac{N2}{N1} = e^{-\frac{h\gamma}{kT}}$

Therefore $\frac{N1}{N2} = e^{\frac{h\gamma}{kT}}$

$$U\gamma = \frac{A_{21}}{B21\left[\frac{B_{12}}{B_{21}}e^{\frac{h\gamma}{kT}} - 1\right]} \quad \rightarrow (4)$$

According to Planck's radiation law equation for $U\gamma$ is given by

$$U\gamma = \frac{8\pi h\gamma 3}{c3\left[e^{\frac{h\gamma}{kT}} - 1\right]} \quad \rightarrow (5)$$

Compare equation (4) & (5) term by term on the basis of positional identity, we get

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\gamma 3}{c^3}$$

And $\frac{B_{12}}{B_{21}} = 1$ (or) $B_{12} = B_{21}$

which implies that the probability of induced absorption is equal to the probability of stimulated emission.

Therefore at thermal equilibrium the equation for energy density is,

$$U\gamma = \frac{A}{B[e^{h\gamma/kT} - 1]}$$

IV. IMPORTANT TERMINOLOGIES

Pumping: The process of exciting atoms from lower energy state to the higher energy state by supplying of energy from an external source is called pumping.

Lasing: It is a process which leads to the emission of stimulated photons after achieving the population inversion.

Active system: The quantum system between those energy levels, the pumping and lasing action takes place.

Laser cavity:

A laser device consists of an active medium bound between two mirrors as shown in figure-2. The mirrors reflect the photons to and fro through the active medium. Thus the two mirrors along with the active medium forms a laser cavity.

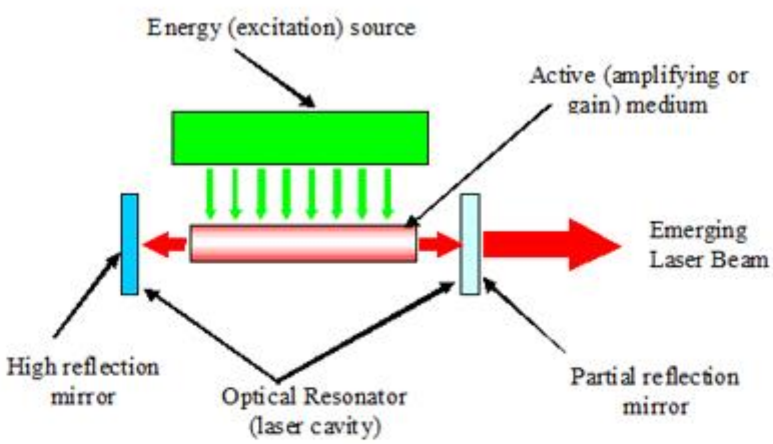


Figure-2

Inside a cavity two types of waves exist, one type comprises of waves moving to the right, and the other one to the left. The two waves interfere constructively if there is no phase difference between the two. But their interference becomes destructive if the phase difference is π. In order to arrange for constructive interference the distance 'L' between the two mirrors should be such that the cavity should support an integral number of half wavelengths.

i.e, $L = (m\lambda/2)$

where, m is an integer > 0

λ is the wavelength of the laser light inside the material of the active medium.

In such a case, a standing wave pattern is established within the cavity and the cavity is said to be resonant at wavelengths hence:

$$\boxed{\lambda = (2L/m)}$$

V. REQUISITES OF A LASER SYSTEM

The three requisites of a laser system are,

a) An excitation source for pumping action,
b) An active medium which supports population inversion, and
c) A laser cavity.

VI. CONDITION FOR LASER ACTION

Population inversion and Metastable state:

Population inversion is the state of a system at which the population of a particular higher energy state is more than that of a specified lower energy state.

In a real physical system, the population inversion conditions don't exist under normal conditions. But it is possible to achieve population inversion condition in certain systems which possess a special kind of excited energy states called *metastable states*. A metastable state is different from any other ordinary excited state.

Consider three energy levels E_1, E_2 and E_3 of a quantum system which are such that $E_3 > E_2 > E_1$ as shown in figure-3. Let E_2 be a metastable state. Let the atoms be excited from E_1 to E_3 state by the supply of appropriate energy from an external source. From the E_3 state the atoms undergo spontaneous downward transitions to E_1 and E_2 states rapidly. In spite of new arrival of atoms, the population of E_1 level cannot increase because of ongoing excitation of atoms to higher energy level. But, since E_2 is a metastable state those atoms which get into that state stay over a very long duration (10^{-3}s) because of which the population of E_2 state increases steadily. Under these conditions a stage will be reached wherein the population of E_2 state overtakes that of E_1 which is known as *population inversion*.

If the above conditions are arranged in a laser cavity, then once the population of E_2 exceeds that of E_1 the stimulated emissions outnumber the spontaneous emissions. Soon stimulated photons all identical with respect to phase, wavelength and direction, grow to a very large number which build up the laser light. Hence the condition for laser action for laser action is achieved by means of population inversion.

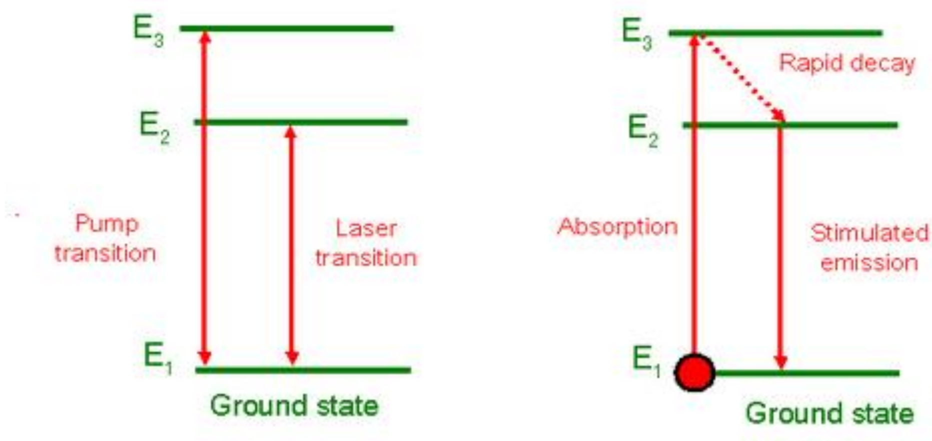


Figure-3

VII. CHARACTERISTICS OF LASER

Laser light has four unique characteristics that differentiate it from ordinary light: these are

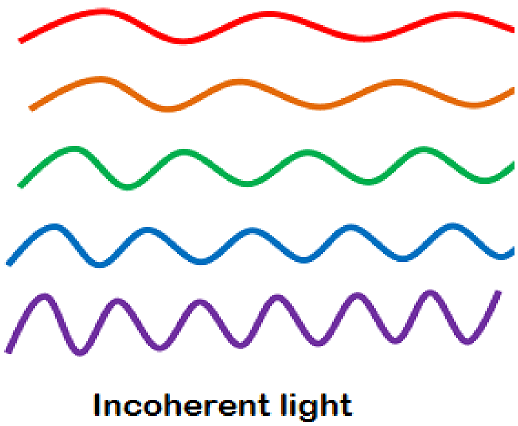
- Coherence
- Directionality
- Monochromaticity
- High intensity

Coherence

We know that visible light is emitted when excited electrons (electrons in higher energy level) jumped into the lower energy level (ground state). The process of electrons moving from higher energy level to lower energy level or lower energy level to higher energy level is called *electron transition*.

In ordinary light sources (lamp, sodium lamp and torch light) the electron transition occurs naturally. In other words, electron transition in ordinary light sources is random in time. The photons emitted from ordinary light sources have different energies, frequencies, wavelengths and colors. Hence the light waves of ordinary light sources have many wavelengths. Therefore, photons emitted by an ordinary light source are out of phase as in figure-5.

In laser, the electron transition occurs artificially. In other words, in laser the electron transition occurs in specific time. All the photons emitted in laser have the same energy, frequency and wavelength. Hence the light waves of laser light have single wavelength or color. Therefore, the wavelengths of the laser light are in phase in space and time as in figure-4. In laser, a technique called *stimulated emission* is used to produce light.



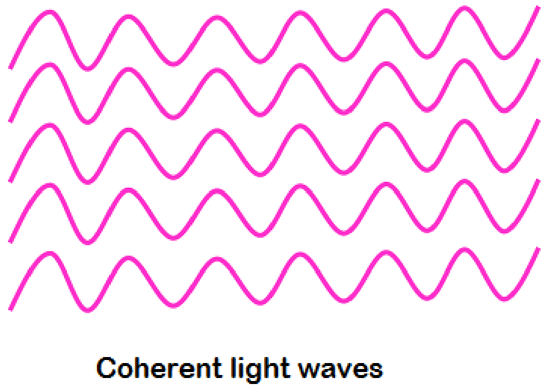


Figure-4 **Figure-5**

Thus light generated by laser is highly coherent. Because of this coherence, a large amount of power can be concentrated in a narrow space.

Directionality

In conventional light sources (lamp, sodium lamp and torchlight), photons will travel in random direction. Therefore, these light sources emit light in all directions as in figure-6.

On the other hand, in laser all photons will travel in same direction. Therefore, laser emits light only in one direction as in figure-6. This is called *directionality of laser light*. The width of a laser beam is extremely narrow. Hence a laser beam can travel to long distances without spreading.

If an ordinary light travels a distance of 2 km, it spreads to about 2 km in diameter. On the other hand, if a laser light travels a distance of 2 km, it spreads to a diameter less than 2 cm.

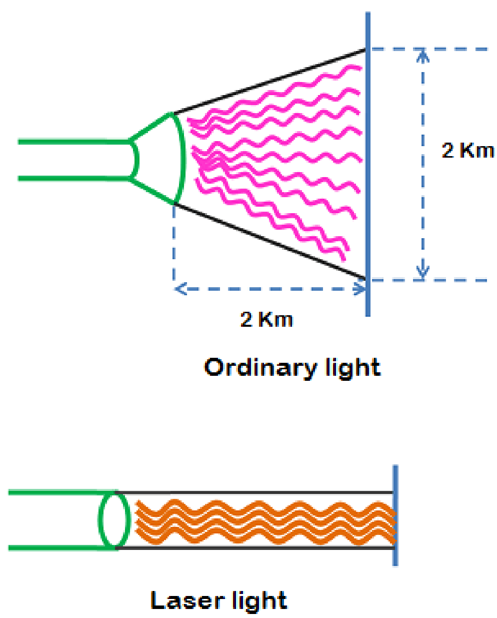


Figure-6

Monochromaticity

Monochromatic light means a light containing a single color or wavelength. The photons emitted from ordinary light sources have different energies, frequencies, wavelengths, or colors. Hence the light waves of ordinary light sources have many wavelengths or colors. Therefore, ordinary light is a mixture of waves having different frequencies or wavelengths.

On the other hand, in laser, all the emitted photons have the same energy, frequency, or wavelength. Hence, the light waves of laser have single wavelength or color. Therefore, laser light covers a very narrow range of frequencies or wavelengths.

High Intensity

You know that the intensity of a wave is the energy per unit time flowing through a unit normal area. In an ordinary light source the light spreads out uniformly in all directions.

If you look at a 100 Watt lamp filament from a distance of 30 cm, the power entering your eye is less than 1/1000 of a Watt.

In laser, the light spreads in small region of space and in a small wavelength range. Hence laser light has greater intensity when compared to the ordinary light.

If you look directly along the beam from a laser, then all the power in the laser would enter your eye. Thus even a 1 Watt laser would appear many thousand times more intense than 100 Watt ordinary lamp.

Thus these four properties of laser beam enable us to cut a huge block of steel by melting. They are also used for recording and reproducing large information on a compact disc (CD).

VIII. NEODYMIUM:YTTRIUM ALUMINUM GARNET LASER

Nd:YAG laser operation was first demonstrated by J.E. Geusic at Bell Laboratories in 1964.

Nd:YAG laser is a neodymium based four level solid state laser. Nd stands for Neodymium (rare earth element) and YAG stands for Yttrium Aluminum Garnet ($Y_3Al_5O_{12}$).

Principle: The active medium Nd:YAG rod is optically pumped by Krypton flash tubes. The Neodymium ions (Nd^{3+}) are raised to excited levels. During the transition from meta stable state to ground state, a laser beam of wavelength 1.064μm is emitted.

Construction:

The construction of Nd:YAG laser is as shown in the below figure-7. A small amount of Yttrium ions (Y^{3+}) is replaced by Neodymium (Nd^{3+}) in the active element of Nd:YAG crystal. The dopant, triply ionized neodymium, Nd(III), typically replaces a small fraction (1%) of the yttrium ions in the host crystal structure of the yttrium aluminum garnet (YAG), since the two ions are of similar size. It is the neodymium ion which provides the lasing activity in the crystal. This active element is cut into a cylindrical rod. The ends of the cylindrical rod are highly polished and they are made optically flat and parallel. This cylindrical rod (laser rod) and a pumping source (flash tube) are placed inside a highly (reflecting) elliptical reflector cavity. The optical resonator is formed by using two external reflecting mirrors. One mirror (M_1) is 100% reflecting while the other mirror (M_2) is partially reflecting.

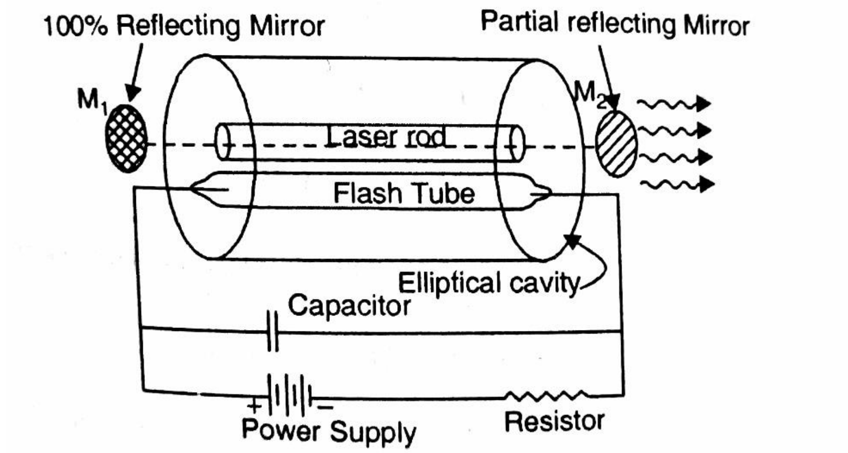


Figure-7

Working: The below figure-8 shows the energy level diagram for Nd:YAG laser. These energy levels are those of Neodymium (Nd^{3+}) ions. When the krypton flash lamp is switched on, by the absorption of light radiation of wavelength 0.73µm and 0.8µm, the Neodymium (Nd^{3+}) atoms are raised from ground level E_0 to upper levels E_3 and E_4 (Pump bands).

The Neodymium ions make a transition from E_3 and E_4 energy levels to E_2 by non-radiative transition. The Neodymium ions are collected in the level E_2 and the population inversion is achieved in E_2 level. E_2 is a metastable state.

An ion makes a spontaneous transition from E_2 to E_1, emitting a photon of energy hγ. This emitted photon will trigger a chain of stimulated photons between E_2 and E_1. The photons thus generated travel back and forth between two mirrors and grow in strength. After some time, the photon number multiplies more rapidly. After enough strength is attained (condition for laser being satisfied), an intense laser light of wavelength 1.06µm is emitted through the partial reflector. It corresponds to the transition from E_2 to E_1.

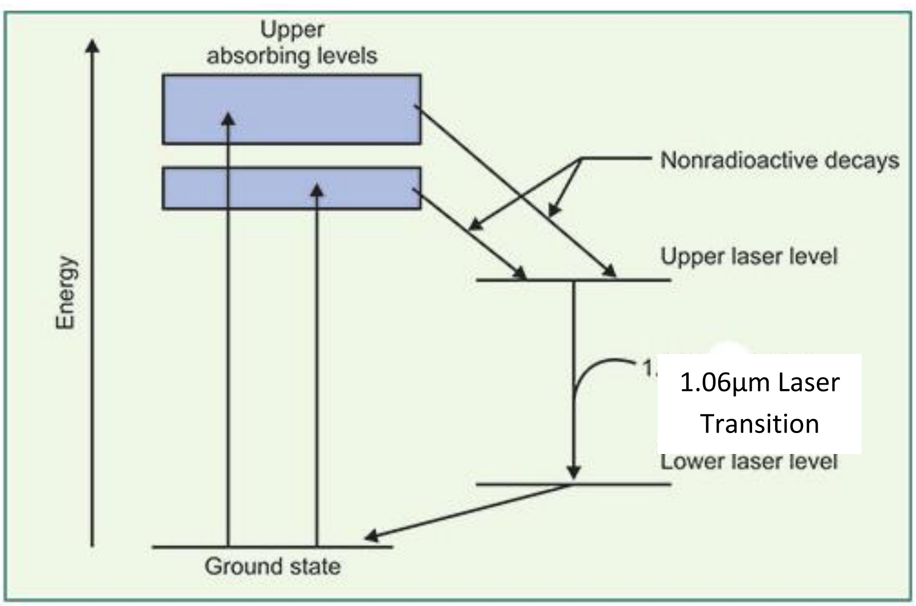


Figure-8

Advantages:

- Low power consumption.
- Offers high gain.
- Good thermal and mechanical properties.
- The efficiency is very high as compared to the ruby laser.

Disadvantage of Nd: YAG Laser

- It is very hard to study the electron energy level structure of Nd^{3+} in YAG.

Applications:

- **Military:** Nd:YAG lasers are used in laser designators and laser rangefinders.

(A laser designator is a laser light source, which is used to target objects for attacking and a laser rangefinder is a rangefinder, which uses a laser light to determine the distance to an object).

- **Medicine:** (1) Used to correct posterior capsular opacification (a condition that may occur after a cataract surgery).
  (2) To remove skin cancers.
- **Manufacturing:** (1) Used for etching or marking a variety of plastics and metals.
  (2) Used for cutting and welding steel.

IX. CARBON DIOXIDE LASER [CO_2 - LASER]

The carbon dioxide laser (CO_2 laser) is one of the most useful gas laser invented and developed by Kumar Patel of Bell Labs in 1964. In CO_2 laser the laser light takes place in the middle IR region within the molecules of carbon dioxide rather than within the atoms of a pure gas therefore CO_2 gas laser is consider a type of molecular gas laser.

Characteristics:

- The CO_2 laser produces a beam of infrared light with the principal wavelength bands centering on 9.4 and 10.6 micrometers.
- The beam divergence of the CO_2 gas laser ranges from 1 to 10 milli radians.
- CO_2 lasers can be operated in either continuous wave (CW) mode or pulse mode. CW mode power output ranges from few watts to over 15000 watts hence it is the highest-power continuous wave lasers that are currently available. In pulse mode the output can be in the millions of watt.

Principle:

CO_2 molecule consists of a central carbon atom with two oxygen atoms attached one on either side. Such a molecule can vibrate in three independent modes of vibrations such as symmetric stretch mode, the bending mode and asymmetric stretch mode.

1. Symmetric Stretch mode

In this mode the carbon atom is stationery and the oxygen atom oscillates simultaneously to and fro along the molecular axis as shown in figure-9.

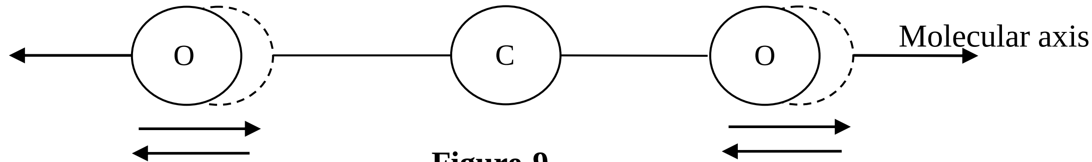


Figure-9

In the energy level diagram C_4 level represents a molecule in pure symmetric mode.

2. Asymmetric stretch mode

In this mode both the oxygen atoms move in one direction while the carbon atom moves in the opposite direction along the molecular axis as shown in the figure-10. The molecule possess high energy in this state of vibration. In energy level diagram C_5 level represents a molecule in its asymmetric mode.

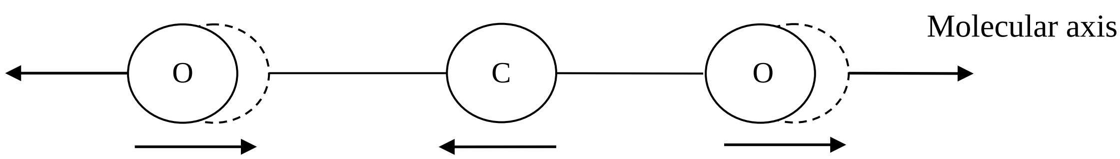


Figure-10

3. Bending mode

In this mode oxygen atoms move perpendicular to the molecular axis as shown in the figure-11. C_2 level is a lowest excited state of the bending mode. In the energy level diagram C_2 level represent the molecule in its bending mode.

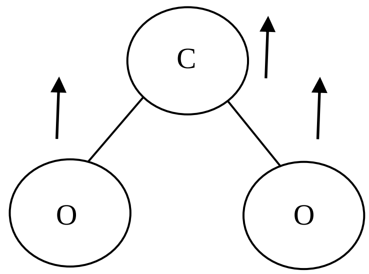


Figure-11

Construction:

Sealed tube design: In this design the gas mixture {10–20% carbon dioxide (CO_2), 10–20% nitrogen (N_2) and the remaining is helium (He)} is sealed in a glass tube and this laser configuration is pumped by RF (Radio Frequency) excitation method as shown in figure-12.

CO_2 laser consists of an active laser medium which is an air cooled gas discharge tube with a total reflector at one end, and an output coupler (a partially reflecting mirror) at the output end. Because CO_2 lasers operate in the infrared the mirrors are silvered, while windows and lenses are made of either germanium or zinc selenide because of their high thermal conductivity and hardness useful for high-power applications.

RF power supply produces the pumping energy, and this energy is coupled to the laser using a transformer designed for maximum power transfer. The coupled output from RF supply is transferred to the electrodes around the gas tube. The nitrogen gas in the gas discharge tube absorbs the pumping energy from the current flow in the gas and transfers the energy through collisions to the CO_2 molecules.

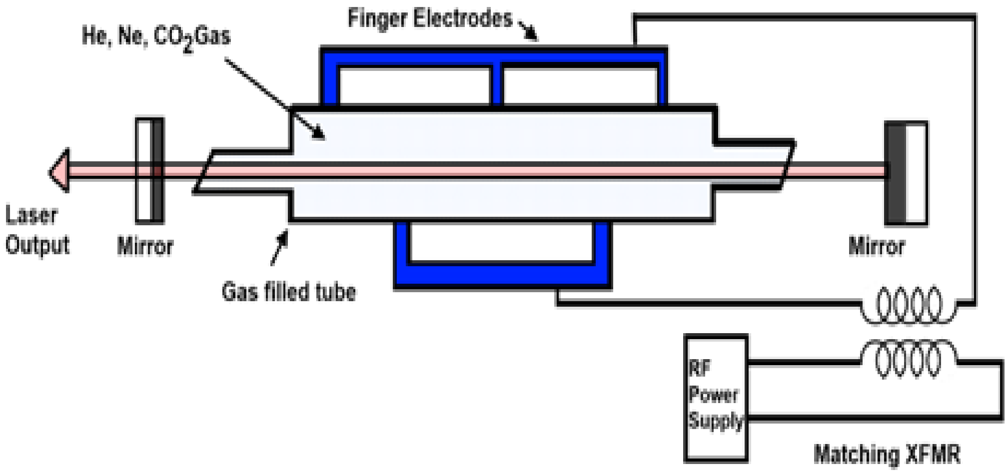


Figure –12

Working (Amplification):

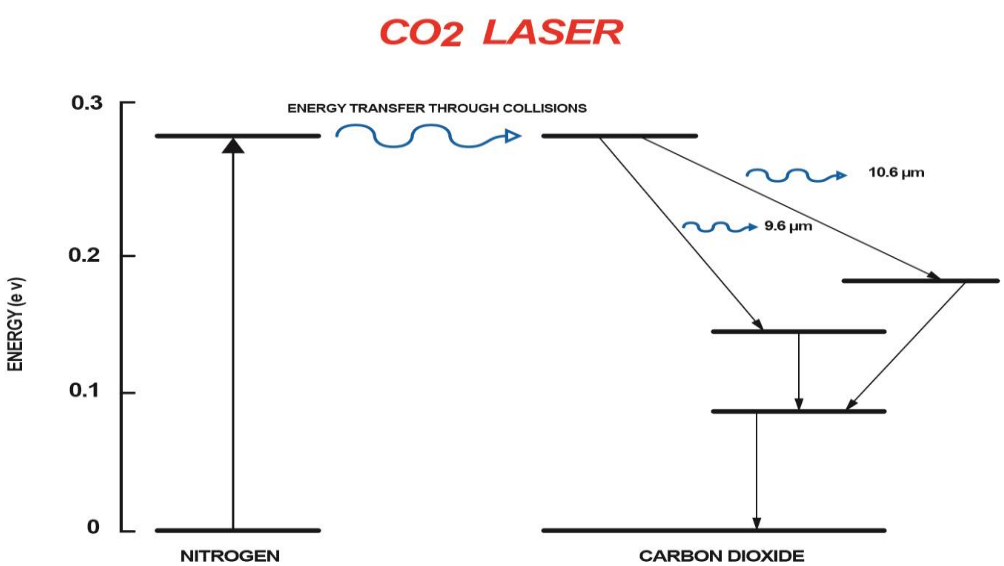


Figure –13

The population inversion in the laser is achieved by the following sequence: electron impact excites vibrational motion of the nitrogen. Because nitrogen is a homonuclear molecule, it cannot lose this energy by photon emission, and its excited vibrational levels are therefore metastable and live for a long time. Now the nitrogen transfers its energy through collision to the carbon dioxide molecule in the (001) asymmetric energy level which causes vibrational excitation of the carbon dioxide, which is sufficient for population inversion. Now the few excited CO_2 molecule drops from the excited level (001) into another symmetric energy level (100) which produces a wavelength of 10.6μm and another few CO_2 molecule drops from the excited level (001) into another bending energy level (020) which produces a wavelength of 9.6μm as shown in figure-13. The transition of the nitrogen molecules to ground state takes place by collision with cold helium atoms which must be constantly cooled in order to sustain the population inversion in the carbon dioxide molecules, and this takes place as the helium atoms strike the walls of the container.

Applications:

- Because of the high power levels available, CO_2 lasers are frequently used in industrial applications for cutting and welding.
- Because water absorbs this frequency of light very well, CO_2 lasers are used in laser surgery and skin resurfacing or laser facelifts.
- Because the atmosphere is quite transparent to infrared light, CO_2 lasers are also used for military range finding using LIDAR techniques.

X. APPLICATIONS OF LASER

By the virtue of their high intensity, high degree of monochromaticity and coherence, lasers find a remarkable application in the field of medicine, military, industry, communications and etc. Some of the applications are: a) LIDAR b) LASIK

LIDAR

Lidar is an acronym of "light detection and ranging" or "laser imaging, detection, and ranging". It is also called as 3-D laser scanning, a special combination of 3-D scanning and laser scanning.

Lidar is a method for determining ranges (variable distance) by targeting an object or a surface with a laser and measuring the time for the reflected light to return to the receiver. It can also be used to make digital 3-D representations of areas on the Earth's surface and ocean bottom of the intertidal and near coastal zone by varying the wavelength of light. It has terrestrial, airborne, and mobile applications.

Lidar is commonly used to make high-resolution maps, with applications in surveying, geodesy, geomatics, archaeology, geography, geology, geomorphology, seismolog y, forestry, atmospheric physics, laser guidance, airborne laser swath mapping (ALSM), and laser altimetry.

Lidar uses ultraviolet, visible, or near infrared light to image objects. It can target a wide range of materials, including non-metallic objects, rocks, rain, chemical compounds, aerosols, clouds and even single molecules. A narrow laser beam can map physical features with very high resolutions; for example, an aircraft can map terrain at 30-centimetre resolution or better.

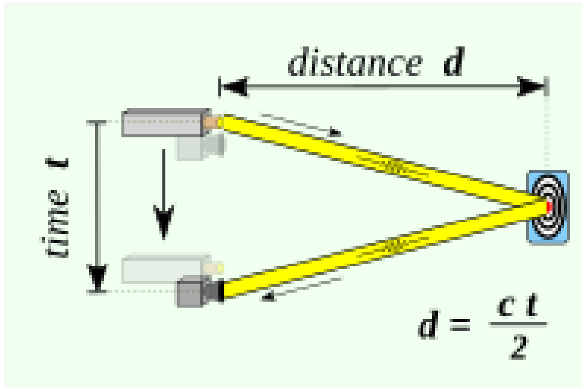


Figure –14

A lidar determines the distance of an object or a surface with the formula:

$$d = c.t/2$$

where c is the speed of light, d is the distance between the detector and the object or surface being detected, and t is the time spent for the laser light to travel to the object or surface being detected, then travel back to the detector.

Types of LiDAR

Functionally, LiDAR systems are either airborne or terrestrial.

Airborne LiDAR

Airborne LiDAR is placed on a drone or helicopter and is helpful for applications that require a bird's eye view of a vast area.

There are two types of standard LiDAR.

The first, topographic, which uses a near-infrared laser to map land areas.

The second, bathymetric, which uses a green water-penetrating light to map underwater terrain.

Terrestrial LiDAR

Terrestrial LiDAR works on the ground and is either mobile or static. Mobile LiDAR systems mount on moving platforms such as autonomous vehicle AI applications to identify objects in the driving environment. Unlike mobile, static LiDAR systems are installed on stationary structures such as tripods—this type of LiDAR is prevalent in archaeology, surveying, mining, and engineering. LiDAR data is accurate, fast, and beneficial for any location where the structure and shape of objects must be determined.

LiDAR is a valuable technology for several industries, from autonomous vehicles to surveying.

Below are some interesting applications of LiDAR:

1. Autonomous Vehicles

   For self-driving vehicle applications, LiDAR provides a longer-range alternative to still image and video cameras, which aren't as effective in poor atmospheric conditions like rain and fog. Vehicles fitted with LiDAR systems collect data such as road markings, traffic signs, pedestrians, road obstructions, and other vehicles. LiDAR is typically used with vision-based systems for fully autonomous (Level 5) vehicles.

2. Aerial Inspection
   Drones/UAV LiDAR data provides valuable aerial insight into industrial assets that are difficult to inspect, including power lines, civil infrastructure, and other industrial assets, to reduce operational maintenance costs.
3. Precision Agriculture
   LiDAR can help agriculture technology (agtech) companies pinpoint areas to optimize water, fertilizer, and herbicides or manage pest control to improve crop yield.
4. Forestry and Land management
   LiDAR can be used to measure the vertical structures and density of the canopy in forests. This data can then be analysed for environmental impact, land management, and fire prevention planning.
5. Survey and mapping
   LiDAR creates accurate maps and digital elevation models for geographic information systems (GIS) to aid civil and commercial surveying and mapping applications.
6. Renewable Energy
   LiDAR can identify requirements for harnessing solar and wind energy, such as optimal solar panel positioning. It can calculate direction and wind speed to allow the operators of wind farms to build and place turbines.
7. Robotics
   LiDAR is used to equip robots with mapping and navigation capabilities. The technology trains an autonomous system to recognize the distance between the vehicle and other objects in the environment.

LASIK

LASIK or Lasik (laser-assisted in situ keratomileusis), commonly referred to as laser eye surgery or laser vision correction, is a type of refractive surgery for the correction of myopia, hyperopia, and an actual cure for astigmatism, since it is in the cornea. LASIK surgery is performed by an ophthalmologist who uses a laser or microkeratome to reshape the eye's cornea in order to improve visual acuity. For most people, LASIK provides a long-lasting alternative to eyeglasses or contact lenses. LASIK is a type of refractive surgery. This kind of surgery uses a laser to treat vision problems caused by refractive errors. You have a refractive error when your eye does not refract (bend) light properly. For you to see clearly, light rays must travel through your cornea and lens. The cornea and lens refract the light so it lands on the retina. The retina turns light into signals that travel to your brain and become images. With refractive errors, the shape of your cornea or lens keeps light from bending properly. When light is not focused on the retina as it should be, your vision is blurry. With LASIK, your ophthalmologist uses a laser to change the shape of your cornea. This laser eye surgery improves the way light rays are focused on the retina. LASIK is used to treat myopia (near sightedness), hyperopia (far sightedness) and astigmatism. The goal of LASIK is to correct refractive error to improve vision. LASIK eye surgery may reduce the need for eyeglasses or contact lenses. In some cases, it may even allow us to do without them completely.

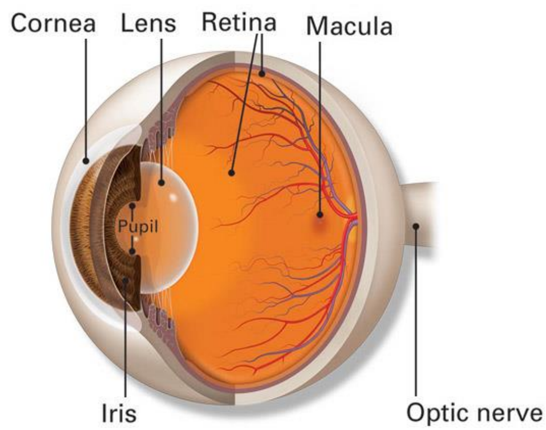


Figure –15